Nahid Hasan

nahidhasan2003131@gmail.com | Google Scholar | GitHub | LinkedIn

Research Interests

Natural Language Processing (NLP) · Multimodal Learning · Low-Resource & Multilingual NLP

Education

Rajshahi University of Engineering & Technology (RUET), Bangladesh Mar 2022 – Aug 2026
B.Sc. in Computer Science and Engineering
CGPA: 3.68/4.00

Research Experience

CAAM-Net: Conflict-Aware Multimodal Bangla Clickbait Detection 2025–2026
Undergraduate Thesis Research · Manuscript Submitted (2026)

- Integrated BanglaBERT and ConvNeXtV2-Tiny via bidirectional cross-attention and conflict-aware prediction for conflicting title–thumbnail cues.
- Designed synthetic conflict training through opposite-class thumbnail swaps; validated 8,108 title–thumbnail pairs and compared 11 baselines using channel-disjoint testing and ablations.
- Achieved mean Macro-F1 of 0.7641 (standard) and 0.7512 (synthetic conflict), exceeding the strongest conventional multimodal baseline by 4.45 and 15.58 percentage points, respectively.

Bangla Hate Speech Detection: Transformer Ensembles & Multi-task Learning 2025
Research leading to BLP-2025 publication

- Developed multilingual transformer models for Bangla hate speech classification across type, target, and severity.
- Designed task-specific transformer ensembles and a shared-encoder multi-task model with separate classification heads.
- Conducted comparative experiments and error analysis involving code-mixed slang, implicit hate, and ambiguous targets.

Publications

Nahid Hasan. “Catalyst at BLP-2025 Task 1: Transformer Ensembles and Multi-task Learning Approaches for Bangla Hate Speech Detection.” *Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025)*, pp. 468–475, Dec. 2025. Association for Computational Linguistics (ACL). DOI: 10.18653/v1/2025.banglalp-1.41.

Selected AI and NLP Projects

AinBondhu: Retrieval-Augmented Legal Question Answering 2026
GitHub · *Python, Flask, PostgreSQL/pgvector, RAG*

- Developed a legal RAG system with Gemini embeddings and PostgreSQL/pgvector, retrieving from 697 indexed statutory sections and producing answers with traceable citations.
- Evaluated 20 queries; retrieved gold evidence in 13 of 14 applicable cases (92.86%) and analyzed retrieval and citation-grounding errors.

AI-Powered Plant Disease Detection and Bilingual Assistant 2025
Hackathon Project · *GitHub* · *TensorFlow Lite, Flask, RAG*

- Collaborated on a lightweight leaf-image disease classifier and a Flask interface with disease-specific care advice.
- Integrated a Bangla–English RAG chatbot with an LLM and web fallback for question answering.

Relevant Industry Experience

Ankabut Software Company Ltd., Rajshahi, Bangladesh Mar 2025
Industrial Attachment Trainee

- Contributed to prototyping an AI chatbot for finding information and services across multiple company websites.
- Supported requirements analysis, testing, and iterative development in an Agile team.

Technical Skills

Programming: Python, C, C++
ML & NLP: PyTorch, Hugging Face Transformers, TensorFlow Lite, scikit-learn, NumPy, pandas
Tools & Databases: SQL, PostgreSQL, pgvector, Flask, Git, LaTeX, Jupyter

Certifications & Professional Development

Aspire Leaders Program Aug 2025
Completed 40 hours of leadership coursework offered by Aspire Institute; among 6,175 finalists selected from 36,050 students.

Data Science Math Skills Feb 2025
Online course offered by Duke University through Coursera, with access provided by FutureNation.

References

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